October 2026

Stephen Kissler, PhD

Assistant Professor of Computer Science, University of Colorado Boulder
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Education

Ph.D. Applied Mathematics, University of Cambridge — 2018
Advisor: Julia Gog
Dissertation: Geographic and demographic transmission patterns of the 2009 A/H1N1 influenza pandemic in the United States

M.S. Applied Mathematics, University of Colorado Boulder — 2014
Advisor: David Bortz
Thesis: Determination of personalized diabetes treatment plans using a two-delay model

B.S. Applied Mathematics, University of Colorado Boulder — 2014
Magna cum laude

Professional Experience

Assistant Professor, University of Colorado Boulder, Boulder CO — Aug 2023 – Present

Affiliate Faculty, Department of Epidemiology, Colorado School of Public Health, Aurora CO — 2026 – Present

Affiliate Faculty, Division of Infectious Diseases, University of Colorado School of Medicine, Aurora CO — 2026 – Present

Postdoctoral Fellow, Harvard T.H. Chan School of Public Health, Boston MA — Jan 2019 – July 2023
- Identified drivers of geographic and temporal variation in antibiotic prescribing in the US
- Produced global COVID-19 projections, measured geographic disparities in COVID-19 transmission in New York City, and characterized SARS-CoV-2 viral load trajectories

Epidemiological Consultant, Partners in Health — Dec 2020 – Apr 2021
- Provided expert consultation on equitable vaccine prioritization strategies leading to guidance documents for public health departments and non-profits in the US

Epidemiological Consultant, National Basketball Association — Jul 2020 – Jun 2021
- Provided model-based guidance on testing, game schedule, and venue occupancy for safely resuming the 2020-21 professional basketball season

Junior Research Fellow, Queens' College, University of Cambridge, Cambridge UK — Oct 2017 – Oct 2020
- Conducted national (UK) citizen science study on human mobility and interaction patterns

Honors and Awards

Kalpana Chawla Outstanding Recent Graduate Award — 2025
Prize by the University of Colorado Alumni Association recognizing alumni who have made exceptional contributions to their field and significant contributions to the community.

NOMIS & Science Young Explorer Award Finalist — 2022
Prize by NOMIS and AAAS for outstanding research performed by the applicant as described in a 1,000-word essay.

Barry R. and Irene Tilenius Bloom Fellowship — 2022

Harvard Chan School trainee award recognizing academic merit, leadership potential, and commitment to improving public health in fields and areas of the world in greatest need.

Institut Mittag-Leffler Fellowship 2018
Awarded funding to attend 2-week residential research exchange on mathematical biology in Stockholm, Sweden.

Research Fellowship, Queens' College, University of Cambridge 2017
Awarded three years of funding to conduct independent research in mathematics.

65th Lindau Nobel Laureate Meeting 2015
One of 650 young scientists selected internationally to attend a week-long academic exchange with 70 Nobel Laureates in physics, chemistry, and medicine & physiology.

Gates Cambridge Scholarship 2014
Merit, leadership, and service-based award for full funding to complete a Ph.D. at the University of Cambridge.

Outstanding Graduate, University of Colorado College of Engineering 2014
Selected for outstanding achievement in research, service, and academics.

Funding

Awarded

Mathematical modeling to accelerate tuberculosis vaccine trials 2026
Co-PI, with Co-PI Tyler Brown (University of Colorado Anschutz)
$75,000
Anschutz-Boulder (AB) Nexus Exploratory Grant

Quantifying the design of effective testing-based policies through infection trajectory modeling 2025-2027
PI, with Co-PI Daniel Larremore (University of Colorado Boulder)
$968,765 to University of Colorado Boulder
NSF, CDC: Mathematical Modeling of Policy Options for Evolving Public Health Challenges

The Watering Whole: Forming a large-scale research collaboration focusing on understanding the modern people, materials, and contaminants flowing with the One Water Cycle 2025-2026
Co-PI, with PI Cresten Mansfeldt and co-PIs Shideh Dashti, Julie Korak, Abbie Liel, Sheldon Masters, Nicole Xu (University of Colorado Boulder)
$100,000 to University of Colorado Boulder
University of Colorado Boulder CRT grants

Submitted/Pending

The timing extremes of acute viral kinetics that drive disease transmission and control 2027-2032
PI
NIH Director's New Innovator Award (DP2); submitted 2026

Reconciling the clinical and epidemiological notions of health 2027
PI
University of Colorado President's Fund for the Humanities; submitted 2026

Invited Presentations

European Centre for Disease Control and Prevention: Modelling Perspectives (Virtual) Jun 2025
Title: Integrating viral kinetics into routine outbreak surveillance

University of Illinois Urbana-Champaign Department of Microbiology Colloquium (Champaign IL) — Apr 2025
Title: Measuring, modeling, and interpreting the kinetics of respiratory viruses

Keynote: Boettcher Foundation Webb-Waring Innovation Summit (Denver CO) — Oct 2024
Title: Innovations in pandemic modeling and beyond

Colorado School of Mines Department of Applied Mathematics and Statistics Colloquium (Golden CO) — Oct 2024
Title: Modeling infectious disease dynamics across scales

Massachusetts Consortium on Pathogen Readiness Variants and Vaccines Meeting (Virtual; Boston MA) — Feb 2024
Title: Viral kinetics of sequential SARS-CoV-2 infections

Predictive Intelligence for Limiting Outbreak Threats (PILOT) workshop (Boston MA) — Sep 2023
Title: Incorporating human behavior into infectious disease models

MIDAS Diversity Conference (Boston MA) — Mar 2023
Title: Introduction to SIR Modeling

Contagion on Complex Social Systems (Boulder CO) — Aug 2022
Title: Linking viral kinetics with disease transmission and control

Expert consultation meeting on the Implementation and Evaluation of Non-Pharmaceutical Interventions (Stockholm) — Jun 2022
Panel title: Non-Pharmaceutical Interventions: The North American Experience

Latin American Symposium on Epidemiological Methods in the COVID-19 Pandemic (Boston MA) — Jun 2022
Title: Introduction to Infectious Disease Models

Modelling Behaviour to Inform Policy for Pandemics (University of Cambridge/Virtual) — Nov 2021
Title: The BBC Pandemic Programme

Fred Hutch/Washington State University Workshop on Virus Dynamics (Virtual) — Oct 2021
Title: Modeling SARS-CoV-2 dynamics across scales

Harvard/Broad Institute Working Group on SARS-CoV-2 Variants (Virtual) — Mar 2021
Title: Viral dynamics of the SARS-CoV-2 variant B.1.1.7

UK Scientific Pandemic Influenza Group on Modelling (SPI-M) (Virtual) — Feb 2021
Title: Viral dynamics of the SARS-CoV-2 variant B.1.1.7

Emory University Theory and Modeling of Living Systems symposium on SARS-CoV-2 modeling (Virtual) — Jan 2021
Title: Preparing for Gen C: Training a new generation of epidemiologists to meet a new generation of public health crises (https://www.youtube.com/watch?v=vjl8ztuo9Ug&t=2540s)

COVID-19 Dynamics and Evolution (Virtual) — Jul 2020
Title: Geographic disparities in SARS-CoV-2 prevalence are predicted by reductions in commuting-style mobility

Council of State and Territorial Epidemiologists training session (Virtual) — Jun 2020
Title: Modeling SARS-CoV-2 transmission in a shifting landscape

Boston University Department of Epidemiology colloquium (Boston MA) — Jan 2020
Title: Accounting for trends in antibiotic prescribing: better prescribing or less disease?

Infectious Disease Dynamics Conference (Ambleside UK) — Sep 2019

Title: Accounting for trends in antibiotic prescribing: better prescribing or less disease?

Probability and Biological Evolution, Centre International de Rencontres Mathématiques (Marseille-Luminy, France) — Jun 2018
Title: Human mobility and strain dynamics

Invited Workshops

Workshop on Public Health Policy, Risk Perception, and Infectious Disease, University of Vermont (Essex VT) — Aug 2026
Invited attendee

Other Presentations

IDWeek (Boston MA) — Oct 2023
Title: Viral kinetics of repeat SARS-CoV-2 infections

IDWeek (Boston MA) — Oct 2023
Title: Assessing the potential reduction in morbidity and mortality attainable by delayed influenza vaccine strain selection

Pre-Research Advising Session (Boulder CO) — Oct 2023
Title: Introduction to infectious disease modeling

Host-Pathogen Supergroup Meeting (Boulder CO) — Oct 2023
Title: Using viral kinetics to infer relationships between SARS-CoV-2 and the immune system

Kern National Network for Flourishing in Medicine 2023 Conference — Sep 2023
Title: Virtue in caring for the individual and caring for the collective

Teaching

University of Colorado Boulder, Boulder CO
- First-year Engineering Seminar (COEN 1830) — Fall 2024, Fall 2025
- Information Theory, Statistical Inference, and Applications (CSCI 7000) — Spring 2024, Spring 2025
- Calculating Biological Quantities (CSCI 2897) — Fall 2023, Fall 2024
- Intro to MS Research (CSCI 5000) – Guest lecturer — Fall 2023

Harvard T.H. Chan School of Public Health, Boston MA
- Introduction to Infectious Disease Dynamics (EPI501) – Guest lecturer — 2021
- Introduction to Clinical Epidemiology (EPI208) – Guest lecturer — 2020 – 2021
- Mathematical Modeling of Infectious Diseases (EPI260) – Teaching fellow — 2020

Boston University, Boston MA
- Advanced Methods in Infectious Disease Epidemiology (BS825) – Guest lecturer — 2021

Brandeis University, Boston MA
- Introduction to Epidemiology, Biostatistics, and Population Health (HSSP 100B) – Guest lecturer — 2021

University of Cambridge, Cambridge UK
- Mathematical Biology (Part II) – Supervisor — 2015 – 2018
- Social Networks and Behavioural Ecology (Part II) – Guest lecturer — 2017

University of Colorado Boulder, Boulder CO

- Technology, Cybersecurity, and Policy seminar (CYBR 5000) – Guest lecturer 2020
- Applied Analysis (APPM 4440) – Course assistant 2012
- Matrix Methods (APPM 3310) – Course assistant 2010

Mentorship

University of Colorado Boulder, Boulder CO

Primary:

L. Hadley: Postdoctoral fellow, Oct 2024-present
M. Richards-Dinger: PhD candidate (applied mathematics), Mar 2024-present
L. Wilson: PhD student (applied mathematics), May 2026-present
M. Kenny: PhD student (IQ Biology), 2026-present
L.X. de Pablo: MS student (computer science), May 2025-present; PhD candidate (Ecology & Evolutionary Biology, co-mentor), Sep 2024-present
P. Simaku: MPH student (Colorado School of Public Health), Sep 2026-present

Alumni:

C. Fulton: PhD candidate (applied mathematics), Jan 2024-2025
N. Rahal-Arabi: MS candidate (computer science), Oct 2024-2025
A. Paul: MS candidate (computer science), Oct 2024-2025
L. Slemp: PhD candidate (Biochemistry), Sep 2024-Dec 2024 (rotation)

Co-mentor:

K. Bardsley: PhD candidate (Ecology & Evolutionary Biology), Sep 2024-present
A. Thilagar: PhD candidate (Computer Science), Sep 2024-May 2025
K. Bubar: PhD candidate (Computer Science), Aug 2023-May 2024
C. Middleton: PhD candidate (Computer Science), Aug 2023-May 2025
I. Kasarskis: PhD candidate (Environmental and Occupational Health), Feb 2024-Mar 2025

Harvard T.H. Chan School of Public Health, Boston MA

Mentored four graduate students on various projects, including predicting the impact of a gonococcal vaccine on antibiotic prescribing, using point-of-care diagnostics to detect pathogens with antimicrobial resistance, and estimating spatial and temporal variation in antibiotic prescribing and antibiotic resistance. 2019 – 2023.

University of Cambridge, Cambridge UK

Mentored a doctoral student on identifying optimal distance metrics for geographic disease transmission models. 2017 – 2019.

Service

General/Field

- Editorial Board Member, Communications Health 2026 – present
- Guest Editor, PLOS Computational Biology 2026
- National Science Foundation Emerging Mathematics in Biology (eMB) review panelist 2025
- Marsden fund application review panelist 2024
- National Science Foundation IHBEM application review panelist 2023
- Boettcher Foundation COVID-19 Innovation Fund review panelist 2020

University of Colorado Boulder

- BioFrontiers Council Member 2024 – present
- Teaching Circle Coordinator Spring 2024
- Graduate Committee Member/Mentorship subgroup lead 2023 – present
- Humanities & Social Sciences Committee Member 2023 – present
- Infectious disease subgroup moderator 2023 – present
- New faculty lunch organizer 2023
- ABNexus grant reviewer 2023

- K. Bubar dissertation defense committee 2023 – present
- C. Middleton dissertation defense committee 2023 – present
- I. Kasarskis dissertation defense committee 2024 – present
- B. Mirphochoev MS thesis defense committee 2024
- IQBiology Group rotation mentor Spring 2024
- M. Moore (Ecology & Evolutionary Biology) thesis committee 2026 – present

Harvard T.H. Chan School of Public Health, Boston MA

- Fungal pathogen modeling working group founding member 2023
- Member of the WHO Access to COVID-19 Tools Accelerator (ACT-A) modeling consortium 2021 – 2023
- Volunteer with Get Out the Vax (bilingual phone banking to register people for vaccination) 2021
- Podcast host for *Pandemic: Coronavirus Edition* 2020 – 2022
- Contributing member of the CDC COVID-19 modeling working group 2020
- Harvard T.H. Chan Postdoctoral Association Departmental Representative (Immunology and Infectious Diseases), Mental Health Initiative Coordinator 2019 – 2022
- Ad-hoc journal reviewer (BMJ Open, Clinical Infectious Diseases, eLife, Elsevier S&T Books, Emerging Infectious Diseases, Epidemics, JAMA Network Open, J. Roy. Soc. Interface, The Lancet, Nature, Nature Communications, Nature Human Behaviour, Phil. Trans. Roy. Soc. B, PLOS Computational Biology, PLOS Medicine, PLOS One, PNAS, Science) 2019 – present
- Guest lecturer for the MIDAS Conference to Increase Diversity in Mathematical Modeling and Public Health 2019

University of Cambridge, Cambridge UK

- Queens' College undergraduate admissions interviewer 2017
- Gates Cambridge Orientation Committee treasurer 2016
- Addenbrookes Hospital chaplaincy volunteer 2016-2018
- Sutton Trust Summer School volunteer and presenter 2016 – 2017
- Worms and Bugs Seminars co-organizer (cross-disciplinary disease modeling seminar) 2015 – 2018
- King's College Graduate Society treasurer 2015 – 2016
- King's College Charitable Giving Stewardship Committee, Graduate Representative 2015 – 2016
- Cambridge Science Festival Maths Open Day volunteer and presenter 2015
- Cambridge Maths Circle Outreach Day volunteer 2014

University of Colorado Boulder, Boulder CO

- Engineering Excellence Fund Board Member/Chair (oversaw $500,000 grant budget) 2011 – 2013
- Society for Industrial and Applied Mathematics Chapter Co-President 2010 – 2014

Bibliography

Peer-reviewed publications

L. Hadley, R.M. Milwid, V. Hongoh, R. Wasfi, **S.M. Kissler**, I. Papst. Making inferences with incomplete epidemiological data: a proof-of-concept estimating measles vaccine coverage across Canada. *Vaccine,* accepted.

K. Bardsley, L.X. de Pablo, E. Keppler Canada, N. Ormaza-Zulueta, Z. Mehrabi, **S.M. Kissler**. Modeling the impact of respiratory disease outbreaks on the United States agricultural workforce. *Communications Health,* accepted.

W. Schaffner, S. Gravenstein, **S. Kissler**, O.A. Cornely, D. Mehta, M. García Cenoz, N. Vicic, D. Rudin. Multicomponent vaccination against seasonal influenza and COVID-19: a strategy to address the continued global burden of disease in adults. *Vaccine,* 93:129082, 2026.

L. Hadley, S. Ben Miled, H.E. Clapham, B.A. Djaafara, L. Dyson, S. Funk, J.R. Gog, H. Heesterbeek, E.M. Hill, E. Howerton, V. Isham, A. Kebir, M. Keeling, **S.M. Kissler**, J. Lessler, K. Leung, E. McBryde, J.M. McCaw, C.E.G. Mercado, D. Mollison, S.M. Thumbi, W. Pan-Ngum, L.

Pellis, F.J. Pérez-Reche, M.J. Plank, R.N. Thompson, C. Tran-Kiem. How understanding the diversity of perspectives and systems in governments can increase the impact of scientific research. *Communications Health,* 1:13, Jul. 2026.

M. Kohli, M. Maschio, A. Lee, **S. Kissler**, S. Carroll, N. van de Velde, E. Beck, O. Balogh, K. Joshi. The cost-effectiveness of vaccination against COVID–19 in at-risk populations and older adults in the United Kingdom: Projections using a dynamic transmission model. *Vaccine* 85:128687, Jul. 2026.

L. Hadley, N. Holliman, K. Xu, E. Bogucka, X. Shu, D. Archambault, K. Landgren, E.M. DeGennaro, **S.M. Kissler**. Visualizing epidemiological models for policy: design principles for effective communication. *Frontiers in Public Health,* 14:1798154, May 2026.

T.S. Brown, K. Nelson, **S. Kissler,** L. Martinez, C.R.Horsburgh, L.F.White, B. Mathema, K.R. Jacobson. Understanding and exploiting superspreading to disrupt *Mycobacterium tuberculosis* transmission. *Lancet Infectious Diseases,* May 2026.

J.A. Hay, D.B. Larremore, A.V. Aatresh, **S.M. Kissler**. Integrating viral kinetics into routine outbreak surveillance: challenges, opportunities, and lessons from COVID-19. *Philosophical Transactions of the Royal Society B,* 20240347, Apr. 2026.

K.N. Nelson, T.S. Brown, **S.M. Kissler**. Diagnostic testing as a public health intervention: challenges and opportunities. *Future Virology,* Dec. 2025.

K.I.O. Roster, **S.M. Kissler**, E. Omoregie, J.C. Wang, H. Amin, S. Di Lonardo, S. Hughes, Y.H. Grad. Surveillance strategies for the detection of new pathogen variants across epidemiological contexts. *PLOS Computational Biology,* Sep. 2024

M. Lipsitch, M.T. Bassett, J.S. Brownstein, P. Elliott, D. Eyre, M.K. Grabowski, J.A. Hay, M. Johansson, **S.M. Kissler,** D.B. Larremore, J. Layden, J. Lessler, R. Lynfield, D. MacCannell, L.C. Madoff, C.J.E. Metcalf, L.A. Meyers, S.K. Orofi, C. Quinn, A.I. Ramos Bento, N. Reich, S. Riley, R. Rosenfeld, M.H. Samore, R. Sampath, R.B. Slayton, D.L. Swerdlow, S. Truelove, J.K. Varma, Y.H. Grad. Infectious disease surveillance needs for the United States: Lessons from COVID-19. *Frontiers in Public Health,* Jul. 2024

M.C. Kline, **S.M. Kissler,** L.K. Whittles, M.L. Barnett, Y.H. Grad. Spatiotemporal trends in Group A Streptococcal pharyngitis in the United States. *Clinical Infectious Diseases,* May 2024.

S.M. Kissler*, K.I.O. Roster*, R. Petherbridge, A. Mehrotra, M.L. Barnett, Y.H. Grad. Drivers of geographic patterns in outpatient antibiotic prescribing in the United States. *Clinical Infectious Diseases,* Mar. 2024. (*denotes equal contribution)

C.G. Tai, M.J. Haviland, **S.M. Kissler**, R.M. Lucia, M. Merson, L.L. Maragakis, D.D. Ho, D.J. Anderson, J. DiFiori, N.D. Grubaugh, Y.H. Grad, C.D. Mack. Low antibody levels associated with significantly increased rate of SARS-CoV-2 infection in a highly vaccinated population from the US National Basketball Association. *Journal of Medical Virology,* 96(3): e29505, Mar. 2024

Andrew Atkeson, **Stephen Kissler.** The impact of vaccines and behavior on US cumulative deaths from COVID-19. *Brookings Papers on Economic Activity,* Mar. 2024

J.W. Olejarz, K.I.O. Roster, **S.M. Kissler**, M. Lipsitch, Y.H. Grad. Optimal environmental testing frequency for outbreak surveillance. *Epidemics,* Feb. 2024

A.D. Wilcock, **S.M. Kissler,** A. Mehrotra, B.E. McGarry, B.D. Sommers, D.C. Grabowski, Y.H. Grad, M.L. Barnett. Clinical risk and outpatient therapy utilization for COVID-19 in the Medicare population. *JAMA Health Forum*, Jan. 2024

S.M. Kissler, J.A. Hay, J.R. Fauver, C. Mack, C.G. Tai, D.J. Anderson D.D. Ho, N.D. Grubaugh, Y.H. Grad. Viral kinetics of sequential SARS-CoV-2 infections. *Nature Communications*, Oct. 2023

H.S. Kwok, A.M. Freedy, A.P. Siegenfeld, J.W. Morriss, A.L. Waterbury, **S.M. Kissler,** B.B. Liau. Drug addiction unveils a repressive methylation ceiling in EZH2-mutant lymphoma. *Nature Chemical Biology*, Mar. 2023

S.M. Kissler, B. Wang, A. Mehrotra, M. Barnett, Y.H. Grad. Impact of respiratory infection and chronic comorbidities on early pediatric antibiotic dispensing in the United States. *Clinical Infectious Diseases,* 76(3): 382-388, Feb. 2023

J.A. Hay*, **S.M. Kissler,*** J.R. Fauver*, C. Mack*, C.G. Tai, R.M. Samant, S. Connolly, D.J. Anderson, G. Khullar, M. MacKay, M. Patel, S. Kelly, A. Manhertz, I. Eiter, D. Salgado, T. Baker, B. Howard, J.T. Dudley. C.E. Mason, M. Nair, Y. Huang, J. DiFiori, N.D. Grubaugh, Y.H. Grad. Quantifying the impact of immune history and variant on SARS-CoV-2 viral kinetics and infection rebound: A retrospective cohort study. *eLife,* 11:e81849, Nov. 2022 (*denotes equal contribution)

C.B. Boyer, E. Rumpler, **S.M. Kissler,** M. Lipsitch. Infectious disease dynamics and restrictions on social gathering size. *Epidemics* 40:100620, Nov. 2022

S.M. Kissler, Y.H. Grad. Anticipating racial/ethnic mortality displacement from COVID-19. *American Journal of Epidemiology,* 191(8): 1519-1520, Aug. 2022

J.A. Watson, **S.M. Kissler,** N.P.J. Day, Y.H. Grad, N.J. White. Characterizing SARS-CoV-2 viral clearance kinetics to improve the design of antiviral pharmacometrics studies. *Antimicrobial agents and chemotherapy,* 66(7): e00192-22, Jul. 2022

S. Lew-Levy, R. Reckin, **S.M. Kissler,** I. Pretelli, A.H. Boyette, A.N. Crittenden, R.V. Hagen, R. Haas, K.L. Kramer, J. Koster, M.J. O'Brien, K. Sonoda, T.A. Surovell, J. Stieglitz, B. Tucker, N. Lavi, K. Ellis-Davies, H.E. Davis. Socioecology shapes child and adolescent time allocation in twelve hunter-gatherer and mixed-subsistence forager societies. *Scientific reports,* 12(1): 8054, May 2022

D.S. Sun, **S.M. Kissler,** S. Kanjilal, S.W. Olesen, M. Lipsitch, Y.H. Grad. Analysis of multiple bacterial species and antibiotic classes reveals large variation in the association between seasonal antibiotic use nad resistance. *PLoS Biology,* 20(3): e3001579, Mar. 2022

S.M. Kissler*, J. Fauver*, C Mack*, ..., N Grubaugh, Y. Grad. Viral dynamics of SARS-CoV-2 variants in vaccinated and unvaccinated individuals. *New England Journal of Medicine,* Dec. 2021 (*denotes equal contribution)

C. Giannitsarou, **S.M. Kissler**, F. Toxvaerd. Waning Immunity and the Second Wave: Some Projections for SARS-CoV-2. *American Economic Review: Insights,* 3(3), Sep. 2021

S.M. Kissler*, J. Fauver*, C Mack*, ..., N Grubaugh, Y.H. Grad. Viral dynamics of acute SARS-CoV-2 infection and applications to diagnostic and public health strategies. *PLOS Biology,* 19(7): e3001333, July 2021 (*denotes equal contribution)

K.C. Ma, T.F. Menkir, **S.M. Kissler**, Y.H. Grad, M. Lipsitch. Modeling the impact of racial and ethnic disparities on COVID-19 epidemic dynamics. *eLife*, 10:e66601, May 2021

S.M. Kissler, M. Klevens, M.L. Barnett, Y.H. Grad. Childhood respiratory outpatient visits correlate with socioeconomic status and drive geographic patterns in antibiotic prescribing. *Journal of Infectious Diseases,* Apr. 2021
Commentary: S.E. Cosgrove, E.Y. Klein. Reassessing the link between healthcare access and outpatient antibiotic prescribing. *Journal of Infectious Diseases,* Apr. 2021

S.M. Kissler*, M. Mitchell*, Y.H. Grad. Reduction in antibiotic prescribing attainable with a gonococcal vaccine. *Clinical Infectious Diseases,* Mar. 2021 (*denotes equal contribution)

D.B. Larremore, B.K. Fosdick, K.M. Bubar, S. Zhang, **S.M. Kissler**, C.J.E. Metcalf, C.O. Buckee, Y.H. Grad. Estimating SARS-CoV-2 seroprevalence and epidemiological parameters with uncertainty from serological surveys. *eLife*, Mar. 2021

K.M. Bubar, K. Reinholt, **S.M. Kissler**, M. Lipsitch, S. Cobey, Y.H. Grad, D. Larremore. Model-informed COVID-19 vaccine prioritization strategies by age and serostatus. *Science,* eabe6959, Jan. 2021

S.M. Kissler*, N. Kishore*, M. Prabhu*, D. Goffman*, Y. Beilin*, R. Landau, C. Gyamfi-Bannerman, B.T. Bateman, J. Snyder, A.S. Razavi, D. Katz, J. Gal, A. Bianco, J. Stone, D. Larremore, C.O. Buckee, Y.H. Grad. Reductions in commuting mobility are correlated with geographic differences in SARS-CoV-2 prevalence in New York City. *Nature Communications,* Aug. 2020 (*denotes equal contribution)

J.A. Firth, J. Hellewell, P. Klepac, **S.M. Kissler**, A.J. Kucharski, L.G. Spurgin, CMMID COVID-19 working group. Combining fine-scale social contact data with epidemic modelling reveals interactions between contact tracing, quarantine, testing and physical distancing for controlling COVID-19. *Nature Medicine,* Aug. 2020

A.J. Kucharski, P. Klepac, A. Conlan, **S.M. Kissler**, M. Tang, H. Fry, J.R. Gog, J. Edmunds, CMMID COVID-19 Working Group. Effectiveness of isolation, testing, contact tracing and physical distancing on reducing transmission of SARS-CoV-2 in different settings. *Lancet Infectious Diseases*, Jun. 2020

A.L. Hicks, **S.M. Kissler**, T.D. Mortimer, K.C. Ma, G. Taiaroa, M. Ashcroft, D.A. Williamson, M. Lipsitch, Y.H. Grad. Targeted surveillance strategies for efficient detection of novel antibiotic resistance variants. *eLife,* 9:e56367, Jun. 2020

S.M. Kissler*, C. Tedijanto*, E. Goldstein, Y.H. Grad, M. Lipsitch. Projecting the transmission dynamics of SARS-CoV-2 through the postpandemic period. *Science*, 368(6493):860-868, May 2020 (*denotes equal contribution)
Info and Metrics: Altmetric score of 19,192; #1 research output of 72,492 in *Science* and #18 research output of 18,494,520 overall.

S.M. Kissler, R.M. Klevens, M.L. Barnett, Y.H. Grad. Distinguishing the roles of antibiotic stewardship and reductions in outpatient visits in generating a five-year decline in antibiotic prescribing. *Clinical Infectious Diseases,* Mar. 2020

S.M. Kissler, C. Viboud, L. Simonsen, O.N. Bjørnstad, B.T. Grenfell, J.R. Gog. Symbolic transfer entropy reveals the age structure of pandemic influenza transmission from high-volume influenza-like illness data. *J. Roy. Soc. Interface*, 17:20190628, Mar. 2020

B. Nichols, **S.M. Kissler**. Ending the HIV epidemic in the United States. *The Lancet HIV*, Mar. 2020

A.L. Hicks, **S.M. Kissler,** M. Lipsitch, Y.H. Grad. Quantifying the surveillance required to sustain genetic marker-based antibiotic resistance diagnostics. *PLOS Biology*, 17(11): e3000547, Oct. 2019

S. Lew-Levy, **S.M. Kissler**, A.H. Boyette, A.N. Crittenden, I.A. Mabullah, B.S. Hewlett. Who teaches children to forage? Exploring the primacy of child-to-child teaching among Hadza and BaYaka Hunter-Gatherers of Tanzania and Congo. *Evolution and Human Behavior,* in press, Aug. 2019

S.M. Kissler, J.R. Gog, C. Viboud, V. Charu, O.N. Bjørnstad, L. Simonsen, B.T. Grenfell. Geographic transmission hubs of the 2009 influenza pandemic in the United States. *Epidemics,* 26:86-94, Oct. 2018

B. Dalziel, **S.M. Kissler,** J. Gog, C. Viboud, O. Bjørnstad, J.E. Metcalf, B.T. Grenfell. Urbanization and humidity shape the intensity of influenza epidemics in U.S. cities. *Science,* 362(6410):75-79, Oct. 2018

P. Klepac, **S.M. Kissler**, J.R. Gog. Contagion! The BBC Four Pandemic – the model behind the documentary. *Epidemics,* 24:49-59, Mar. 2018

V. Charu, S. Zeger, J.R. Gog, O.N. Bjørnstad, **S.M. Kissler**, L. Simonsen, B.T. Grenfell, C. Viboud. Human mobility and the spatial transmission of influenza in the United States. *PLOS Comp. Bio.,* 13(2):e1005382, Feb. 2017

S.M. Kissler, C. Cichowitz, S. Sankaranarayanan, D.M. Bortz. Determination of personalized diabetes treatment plans using a two-delay model. *J. Theor. Biol.,* 359:101-111, Oct. 2014

Conference proceedings.

M. Kohli, M. Maschio, K. Fust, A. Lee, K. Joshi, **S. Kissler**, N. Van de Velde, E. Beck. EPH90 The potential clinical impact of the Moderna next-generation COVID-19 mRNA-1283 vaccine in United States (US) adults. *Value in Health,* 28(6): S176, Jul. 2025

A. Wilcock, **S. Kissler**, Y. Grad, D. Grabowski, B.E. McGarry, A. Mehrotra, M. Barnett, B. Sommers. Clinical risk and utilization of outpatient antiviral and antibody therapy for COVID-19 in Medicare. *ASHEcon 2024 Conference,* Jun. 2024

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